

Efficient approximation of flow problems with multiple scales in time

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In this article we address flow problems that carry a multiscale character in time. In particular we consider Navier-Stokes flow in a channel on a fast scale that influences the movement of the boundary which undergoes a deformation on a slow scale in time. We derive an averaging scheme that is of first order with respect to the ratio of time-scales α . In order to cope with the problem of unknown initial data for the fast scale problem, we assume near-periodicity in time. Moreover, we construct a second-order accurate time discretisation scheme and derive a complete error analysis for a corresponding simplified ODE system. The resulting multiscale scheme does not ask for the continuous simulation of the fast scale variable and shows powerful speed-ups up to 1:10 000 compared to a resolved simulation. Finally, we present some numerical examples for the full Navier-Stokes system to illustrate the convergence and performance of the approach.

1 Introduction

We are interested in the numerical approximation and long-term simulation of flow problems that carry a multiscale character in time. Such problems appear for example in the formation of atherosclerotic plaque in arteries, where flow dynamics acting on a scale of milliseconds to seconds have an effect on plaque growth in the vessel, which typically takes place within a range of several months. Another application is the investigation of chemical flows in pipelines, where long-time effects of weathering, accelerated by the transported substances, cause material alteration.

These examples have in common, that it is computationally infeasible to resolve the small scale over the whole time interval of interest. In the case of atherosclerotic plaque growth, a suitable time-step of $1/20$ s would require nearly 10^9 steps to cover the period of interest, which is around 6 months.

Inspired by the temporal dynamics of atherosclerotic plaque growth, we will consider the flow in a channel whose boundary is deformed over a long time scale in this work. This deformation is controlled by some concentration $\kappa(t)$ that is governed by a simple reaction equation and depends on the fluid-forces

$$\begin{aligned} \mathbf{v}(0) = \mathbf{v}_0, \quad \nabla \cdot \mathbf{v}, \quad \rho(\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}) - \operatorname{div} \boldsymbol{\sigma}(\mathbf{v}, p) &= \mathbf{f} & \text{in } \Omega(\kappa(t)) \\ \kappa(0) = 0, \quad \kappa' &= R(\mathbf{v}, \kappa). \end{aligned} \tag{1}$$

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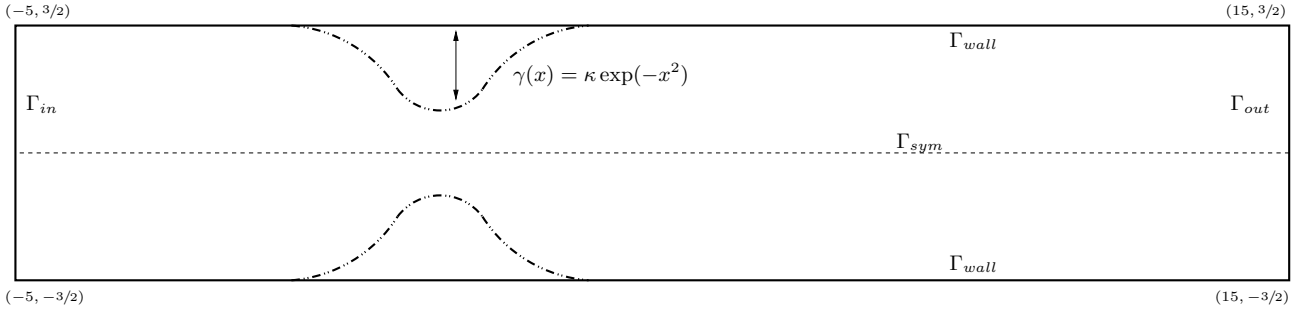


Figure 1: Configuration of the test case. We study flow in a channel with a boundary Γ that depends on a concentration variable κ . This κ follows a simple reaction law with a right-hand side depending on the wall shear stress on Γ .

Here, ρ is the density of the fluid, $\boldsymbol{\sigma} = \rho\nu(\nabla\mathbf{v} + \nabla\mathbf{v}^T) - pI$ the Cauchy stress with the kinematic viscosity ν and $R(\mathbf{v}, \kappa) \geq 0$ a reaction term describing the influence of the fluid dynamical forces (namely the wall shear stress) on the boundary growth. The domain $\Omega = \Omega(\kappa(t))$ depends explicitly on the concentration $\kappa(t)$. We show a sketch of the configuration in Figure 1. The flow problem is driven by a periodic oscillating inflow profile of period 1 s

$$\mathbf{v}^D(t) = \mathbf{v}^D(t + 1s).$$

This period describes the *fast scale* of the problem. While the problem itself is strongly simplified compared to the detailed non-linear mechano-chemical FSI model of plaque-growth [6, 40, 12], we choose the parameters in such a way that the temporal dynamics are very similar.

A challenge for the construction of efficient averaging schemes is that the initial data is typically unknown on the fast scale. Therefore, we make the essential assumption that for a frozen value of the slow variable $\bar{\kappa} = \kappa(t)$ a periodic fast-scale solution exists. For the Navier-Stokes equations this is justified theoretically for small data [25, 14].

While multiscale problems in space are extensively studied in the literature, see e.g. [8, 29], much less works can be found for problems with multiscale character in time. A few works exist that use a homogenisation approach based on asymptotic expansions in time for viscoelastic, viscoplastic or elasto-viscoplastic solids [16, 42, 2, 18]. Under suitable assumptions, the short-scale part of the multiscale algorithm becomes stationary for this class of equations, such that difficulties to define initial values on the short scale are avoided.

If the time scales are close enough that the short-scale dynamics can be resolved within one time step of the long-scale discretisation, the *Variational Multiscale Method* [21, 5] or approaches that construct long-scale basis functions from the short-scale information [30, 1] are applicable. Similar algorithms are also used to construct parallel-in time integrators, for example the *parareal* method [27]. In this work, we are interested in problems with a stronger *scale separation*, where the resolution of the short scale within a long-scale interval is very costly up to computationally unfeasible.

Only very few numerical works can be found concerning flow problems with multiple scales in time. An exception are the works of Masud & Khurram [28, 24], where the *Variational Multiscale Method* is applied, assuming again that the time scales are sufficiently close. On the other hand, several theoretical works exist that show convergence towards averaged equations for specific flow configurations in the situation that the ratio of time scales $\alpha = \frac{t_{\text{fast}}}{T_{\text{long}}}$ tends to zero, see e.g. [22, 7, 26], however without considering practical numerical algorithms or discretisation.

A common numerical approach is to replace the fast problem by an averaged one using a fixed-in-time inflow profile [40, 6]. It is however widely accepted and also confirmed in numerical studies [12] that such a simple averaging does not necessarily reproduce the correct dynamics. In [12] we presented a first multiscale scheme for the approximation of such a problem, however with a focus on the modelling of a full closure of the channel and without any analysis on the robustness and accuracy. Similar algorithms can be found in Sanders et al [34] and by Crouch & Oskay [9] in different applications. In this work, we will derive an improved algorithm in a mathematically rigorous way, including a detailed error analysis for both modelling and discretisation errors. To our knowledge this is the first time that the interplay between modelling errors of the temporal multiscale scheme and temporal discretisation errors on both scales is analysed.

The structure of this article is as follows: In Section 2 we start by introducing a typical temporal multiscale problem involving partial differential equations and discuss the role of the scale separation. Then, in Section 3 we derive effective long term equations that will be the basis for an efficient multiscale approach. Here, we also give a numerical analysis. This however is based on a strongly simplified case considering the coupling of two ODEs. In Section 4 we describe the temporal discretisation of the multiscale scheme and show optimal order convergence in all discretisation parameters: mesh size h , time step size k for the fast problem and time step size K for the slowly evolving variable. In Section 5 we apply the multiscale scheme to the complex problem introduced in Section 2, which is based on the Navier-Stokes equations. We show numerically optimal-order convergence in agreement to the theoretical findings for the simplified system. We conclude with a short summary and a discussion of some open problems.

2 Time scales

2.1 Configuration of the model problem

A sketch of the model problem under consideration is given in Figure 1. On the fast scale we consider a Navier-Stokes flow in a channel whose width depends on a *slowly evolving variable* $\kappa(t)$. The variable domain describing the channel is given by

$$\Omega(\kappa) = \{(x, y) \in \mathbb{R}^2 : -5 \text{ cm} < x < 10 \text{ cm}, |y| < (1.5 - \kappa\gamma(x)) \text{ cm}\}, \quad \gamma(x) = \exp(-x^2). \quad (2)$$

Instead of a complex growth model for the plaque formation as introduced in [40] we use an explicit dependence of the domain on the scalar $\kappa(t)$.

Problem (1) can be formulated on a reference domain $\Omega := \Omega(0)$ by means of an *Arbitrary Lagrangian Eulerian* approach using the map [32]

$$T : \Omega(0) \rightarrow \Omega(\kappa), \quad T(x, y, \kappa) = \begin{pmatrix} x \\ \frac{1.5 - \kappa\gamma(x)}{1.5} y \end{pmatrix} \quad (3)$$

with derivative and determinant given by

$$\mathbf{F} := \hat{\nabla} T = \begin{pmatrix} 1 & 0 \\ -\frac{\kappa\gamma'(x)}{1.5} y & \frac{1.5 - \kappa\gamma(x)}{1.5} \end{pmatrix}, \quad J := \det(\mathbf{F}) = \frac{1.5 - \kappa\gamma'(x)}{1.5}. \quad (4)$$

A 3d configuration could be realised by rotational symmetry. The flow is driven by a periodic inflow profile $\mathbf{v}^D$ on the left boundary with period $t_{fast} \approx 1$ s. We use the period 1 s throughout this paper. The extension to general periods is, however, straightforward.

For the model problem a reaction term, that mimics the long-term dynamics as well as the influence of the wall shear stress on the growth rate of the plaque is given by

$$R(\mathbf{v}, \kappa, t) = \frac{\alpha}{(1 + \kappa)(1 + |\sigma_{WSS}[\mathbf{v}](t)|^2)} \quad \sigma_{WSS}[\mathbf{v}] = \sigma_0^{-1} \int_{\Gamma} \rho \nu [I_d - \vec{n} \vec{n}^T] (\nabla \mathbf{v} + \nabla \mathbf{v}^T) \vec{n} \, d\sigma. \quad (5)$$

Here $\alpha, \sigma_0 > 0$ are problem parameters. The reaction term has been chosen in such a way that a saturation effect is included. The parameter σ_0 will be tuned to give $|\sigma_{WSS}[\mathbf{v}]| = \mathcal{O}(1)$. The parameter α will control the separation of the time scales, a discussion follows in the next section. This model is a simplification of the detailed model considered in [40, 12]. The correct model and a full comprehension of shear effects on plaque formation and growth are still under active discussion. It is however understood that regions of (relatively) low shear stress that exhibit an oscillatory character are more prone to plaque growth [37, 10]. The specific reaction term (5) mimics this behaviour. Moreover, the dependence on the flow problem by means of the wall shear stress is nonlinear and cannot be considered by a simple averaging as done in [39, 40, 41].

2.2 Assumptions on the time scale

In this section we want to specify what we mean by a separation of the problem into the *slowly evolving variable* $\kappa(t)$ and the *fast evolving variables* $(\mathbf{v}(t), p(t))$. The fast scale of the flow problem is defined via the period of the inflow profile $\mathbf{v}^D$, i.e. $t_{fast} = 1s$. We will formulate all the quantities in this unit, i.e. $t_{fast} = \mathcal{O}(1)$. The slow scale is the time scale of interest for the plaque growth, i.e. the boundary movement in our model problem, which is $T_{long} = \mathcal{O}(1 \text{ month}) = \mathcal{O}(10^6 s)$. We assume that the channel width is reduced to approximately half its original size within a time frame of $T \approx T_{long}$, i.e. $\kappa(T_{long}) = \mathcal{O}(1)$. To rescale the problem we introduce the scale parameter

$$\alpha = \frac{t_{fast}}{T_{long}} \approx 10^{-6},$$

which is part of the reaction term $R(\mathbf{v}, \kappa)$ in (5).

Assumption 1 (Properties of the flow problem). *Let $\kappa \in C^1(I)$ for $I = [0, T_{long}]$ with $0 \leq \kappa \leq \kappa_{max} = \mathcal{O}(1)$. We assume that the Navier-Stokes equations on the moving domain $\Omega(\kappa(t))$ admit a solution $(\mathbf{v}(t), p(t))$ satisfying the bound*

$$\|\mathbf{v}(t)\|_{H^2(\Omega(\kappa(t)))} + \|p(t)\|_{H^1(\Omega(\kappa(t)))} \leq c_f, \quad (6)$$

where $c_f = \mathcal{O}(1)$ depends on $\Omega(\kappa(t))$. Further, we assume that $\sigma_0 > 0$ is such that

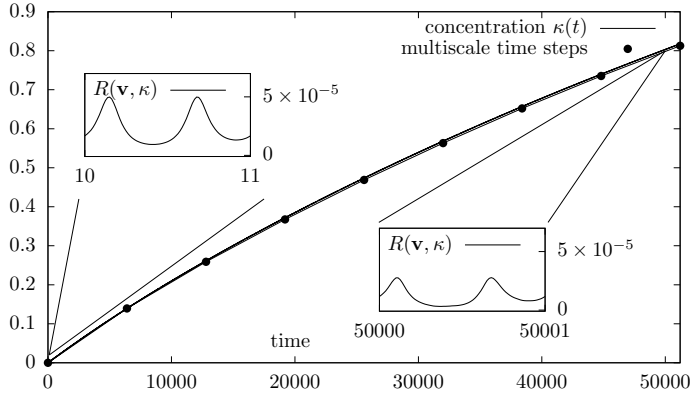
$$|\sigma_{WSS}(\mathbf{v})|^2 \leq c_f \text{ and } R(\mathbf{v}, \kappa) \leq \alpha c_R, \quad (7)$$

with a constant $c_R = \mathcal{O}(1)$.

By (7) we obtain

$$0 \leq \kappa' \leq \alpha c_R \quad (8)$$

showing the slowness of the long term dynamics. The validity of this assumption depends on the Reynolds number and on the regularity of the domain, which is not an issue in our case as long as $\kappa \leq \kappa_{max} < 1.5$ prevents a closure of the channel.



Aim of the multiscale approach is to derive a discretisation scheme that is able to predict the concentration κ based on very large time steps marked by the bullets (here $K = 6\,400\text{s}$). The short term contribution significantly changes from $T = 10\text{s}$ to $T = 50\,000\text{s}$ and initial values for the fast variable $\mathbf{v}$ are not available in a multiscale scheme.

3 Temporal multiscale algorithms

Based on the assumptions made in the previous section, we can now derive a temporal multiscale algorithm. Therefore, we introduce a time-stepping procedure on the slow scale with step-size $K \approx 1$ day. We will use capital letters whenever the slow scale is considered and small letters to specify the fast time scale. A sketch of the configuration is shown in Figure 2.

- We discretise the period of interest on the slow scale in steps $0 = T_0 \rightarrow T_1 \rightarrow \dots \rightarrow T_N = T$ and define a time-stepping scheme to forward the long-scale variable $\kappa(T_n) \mapsto \kappa(T_{n+1})$. The time-step size $K = T_n - T_{n-1}$ is chosen such that it is able to represent the slow dynamics, e.g. $K \approx 1$ day.
- As $\mathcal{O}(1 \text{ day}) = K \gg t_{\text{fast}} = \mathcal{O}(1 \text{ s})$ we can not simulate the fast scale dynamics on the complete interval $[T_{n-1}, T_n]$. Instead we derive an effective long scale model that involves only the local contributions of a periodic fast-scale problem approximating the dynamics in the interval $[T_n - 1 \text{ s}, T_n]$.
- We discretise the fast scale problem on the interval $[T_n - 1 \text{ s}, T_n]$ with a small step size $k \ll K$ and define a suitable time-stepping scheme. Therefore, we fix the slow variable $\kappa = \kappa(T_n)$ and impose periodic boundary conditions in time for the flow problem. We denote the local solution by $(\mathbf{v}^{\kappa(T_n)}(t), p^{\kappa(T_n)}(t))$.
- Finally, we step forward the slow scale variable $\kappa(T_{n-1}) \mapsto \kappa(T_n)$ based on the effective model and the short-scale solutions $(\mathbf{v}^{\kappa(T_n)}(t), p^{\kappa(T_n)}(t))$.

Figure 2 reveals the fundamental difficulty in temporal multiscale approximations: as the fast-scale solution is not computed on the complete intervals $[T_{n-1}, T_n]$ there are no initial values available to compute $\mathbf{v}^{\kappa(T_n)}(t)$. As mentioned above, we make the crucial assumption that the fast scale problem will allow for a unique periodic solution for a fixed slow scale variable κ to cope with this problem. The local fast scale influences will be based on this periodic solution such that systematic errors coming from unknown initial states are avoided.

3.1 Derivation of an effective equation

We aim at deriving an effective equation that describes the slow dynamics of the variable $\kappa(t)$ averaged over cycles of length $t_{fast} = 1$ s. For this purpose we introduce the averaged variable

$$\hat{\kappa}(t) = \int_t^{t+1} \kappa(s) \, ds.$$

By inserting $\pm R(\mathbf{v}(s), \hat{\kappa}(t))$, we have (compare (1))

$$\hat{\kappa}'(t) = \int_t^{t+1} R(\mathbf{v}(s), \kappa(s)) \, ds = \int_t^{t+1} R(\mathbf{v}(s), \hat{\kappa}(t)) \, ds - \int_t^{t+1} (R(\mathbf{v}(s), \hat{\kappa}(t)) - R(\mathbf{v}(s), \kappa(s))) \, ds.$$

Lemma 2 (Averaging error). *Let $\kappa \in C^1([0, T_{long}])$ and $(\mathbf{v}, p)$ be the corresponding solution to the Navier-Stokes problem on $\Omega(\kappa)$ satisfying Assumption 1. It holds that*

$$\max_t \left| \int_t^{t+1} (R(\mathbf{v}(s), \hat{\kappa}(t)) - R(\mathbf{v}(s), \kappa(s))) \, ds \right| \leq c\alpha^2,$$

with a constant $c = c(c_f, c_R)$.

Proof. It holds

$$|R(\mathbf{v}(s), \hat{\kappa}(t)) - R(\mathbf{v}(s), \kappa(s))| = \left| \alpha \frac{\kappa(s) - \hat{\kappa}(t)}{(1 + |\sigma_{WSS}[\mathbf{v}](t)|^2)(1 + \kappa(s))(1 + \hat{\kappa}(t))} \right| \leq \alpha |\kappa(s) - \hat{\kappa}(t)|. \quad (9)$$

We estimate

$$\int_t^{t+1} |\kappa(s) - \hat{\kappa}(t)| \, ds = \int_t^{t+1} \left| \int_t^{t+1} (\kappa(s) - \kappa(r)) \, dr \right| \, ds = \int_t^{t+1} \left| \int_t^{t+1} \int_r^s \kappa'(x) \, dx \, dr \right| \, ds \leq c\alpha, \quad (10)$$

such that a combination of (9) and (10) shows the assertion. $\square$

We can thus approximate the averaged evolution equation for $\hat{\kappa}$ by

$$\hat{\kappa}'(t) = \int_t^{t+1} R(\mathbf{v}(s), \hat{\kappa}(t)) \, ds + \mathcal{O}(\alpha^2). \quad (11)$$

Due to the nonlinearity of $R(\cdot, \cdot)$, the fine scale variations in $\mathbf{v}(s)$ can not simply be averaged. This is the fundamental dilemma of temporal multiscale methods: Equation (11) can not be used for an efficient long time-step discretisation $T_n \rightarrow T_{n+1} = T_n + K$, as it would require knowledge of the solution $\mathbf{v}(T_{n+1})$ which dynamically evolves from $\mathbf{v}(T_n)$ on the fast scale k . At the heart of this problem is the lack of initial values for a local reconstruction of the fast scale problem at a given discrete long time step $T_n \rightarrow T_{n+1}$. To overcome this problem we introduce the periodic-in-time solution $(\mathbf{v}^\kappa, p^\kappa)$ to the Navier-Stokes equation for every fixed parameter $\kappa = \hat{\kappa}(t)$:

$$\nabla \cdot \mathbf{v}^\kappa = 0, \quad \rho(\partial_t \mathbf{v}^\kappa + (\mathbf{v}^\kappa \cdot \nabla) \mathbf{v}^\kappa) - \operatorname{div} \boldsymbol{\sigma}(\mathbf{v}^\kappa, p^\kappa) = \mathbf{f} \quad \text{in } [0, 1] \times \Omega(\kappa) \quad (12)$$

$$\mathbf{v}^\kappa = \mathbf{v}^D \quad \text{on } [0, 1] \times \partial\Omega(\kappa) \quad (13)$$

$$\mathbf{v}^\kappa(1) = \mathbf{v}^\kappa(0) \quad \text{in } \Omega(\kappa) \quad (14)$$

Assumption 3 (Periodic solution to the Navier-Stokes equations). *Let $\kappa \in \mathbb{R}$ with $0 \leq \kappa \leq \kappa_{max}$. We assume that there exists a unique periodic solution $(\mathbf{v}^\kappa, p^\kappa)$ to the incompressible Navier-Stokes equations (12) satisfying the bound*

$$\|\mathbf{v}^\kappa(t)\|_{H^2(\Omega(\kappa(t)))} + \|p^\kappa(t)\|_{H^1(\Omega(\kappa(t)))} \leq c_f \quad (15)$$

for all times $t \in [0, T]$.

The unique existence of such a periodic solution is only guaranteed for small problem data, see [13, 14, 25]. These results will most likely not apply to the higher Reynolds number regime of typical blood flow configurations.

The benefit of the periodic solutions lies in a localisation of the fast scale influences. Given an approximation $\bar{\kappa}_n$ at time T_n , the periodic-in-time solution $(\mathbf{v}^{\bar{\kappa}_n}, p^{\bar{\kappa}_n})$ can be determined without any initial values such that time-stepping $\bar{\kappa}_n \mapsto \bar{\kappa}_{n+1}$ is possible.

We approximate the averaged equation (11) using the periodic solution $\mathbf{v}^{\hat{\kappa}(t)}(s)$ around the frozen value $\hat{\kappa}(t)$

$$\hat{\kappa}'(t) = \int_t^{t+1} R(\mathbf{v}^{\hat{\kappa}(t)}(s), \hat{\kappa}(t)) ds + \int_t^{t+1} (R(\mathbf{v}(s), \hat{\kappa}(t)) - R(\mathbf{v}^{\hat{\kappa}(t)}(s), \hat{\kappa}(t))) ds + \mathcal{O}(\alpha^2). \quad (16)$$

We will show that the second remainder

$$\max_t \left| \int_t^{t+1} (R(\mathbf{v}(s), \hat{\kappa}(t)) - R(\mathbf{v}^{\hat{\kappa}(t)}(s), \hat{\kappa}(t))) ds \right| \quad (17)$$

is small of order $\mathcal{O}(\alpha^2)$. The corresponding analysis is done for a strongly simplified system of equations and presented in the following section. Results for the Navier-Stokes equations would have to deal with the possible non-uniqueness of periodic solutions $\mathbf{v}^{\hat{\kappa}(t)}(s)$.

Having shown that the average $\hat{\kappa}(t)$ satisfies the equation

$$\hat{\kappa}'(t) = \int_t^{t+1} R(\mathbf{v}^{\hat{\kappa}(t)}(s), \hat{\kappa}(t)) ds + \mathcal{O}(\alpha^2)$$

we introduce the solution $\bar{\kappa}(t)$ of the effective equation

$$\bar{\kappa}'(t) = \int_t^{t+1} R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) ds, \quad \bar{\kappa}(0) = \kappa_0. \quad (18)$$

We note that the initial values $\bar{\kappa}(0) = \kappa_0$ and $\hat{\kappa}(0) = \int_0^1 \kappa(t) dt$ do not necessarily coincide. Instead, it holds

$$\hat{\kappa}(0) = \int_0^1 \kappa(t) dt = \kappa_0 + \int_0^1 \int_0^s \kappa'(s) ds dt = \kappa_0 + \mathcal{O}(\alpha). \quad (19)$$

3.2 Analysis of the approximations

In this section, we outline the ideas for showing convergence of the multiscale scheme introduced at the beginning of this section. As mentioned above the analysis for the Navier-Stokes-ODE system is beyond the scope of this work. As periodic solutions $(\mathbf{v}^{\hat{\kappa}(t)}(s), p^{\hat{\kappa}(t)}(s))$ are not necessarily unique, closeness to the dynamic solution $(\mathbf{v}(s), p(s))$ could only be shown for very small problem data. Here,

we simplify the Navier-Stokes equations to a simple ordinary differential equation and consider the coupled problem

$$\kappa'(t) = R(\mathbf{v}(t), \kappa(t)) := \frac{\alpha}{(1 + \kappa(t))(1 + \mathbf{v}(t)^2)} \quad \kappa(0) = \kappa_0, \quad (20a)$$

$$v'(t) + A(\kappa(t))\mathbf{v}(t) = f(t) \quad \mathbf{v}(0) = \mathbf{v}_0. \quad (20b)$$

where $f \in C([0, T])$ is periodic with period 1 and $A(\kappa) \geq A_{\min} > 0$ is a scalar quantity such that the map $\kappa \mapsto A(\kappa)$ is differentiable with a bounded derivative

$$\left| \frac{dA}{d\kappa} \right| \leq c_A \quad (21)$$

for $0 \leq \kappa \leq \kappa_{\max}$. As in the case of the Navier-Stokes coupling it holds $0 \leq R(v, \kappa) \leq c_R \alpha$.

Remark 4 (Discussion of the simplifications). *If we linearise the Navier-Stokes equations by omitting the convective terms $(\mathbf{v} \cdot \nabla)\mathbf{v}$ as well as the domain convection coming from the ALE transformation, the problem reduces to the Stokes equations transformed to a fixed reference domain*

$$\operatorname{div}(J\mathbf{F}^{-1}\mathbf{v}) = 0, \quad J\partial_t\mathbf{v} - \operatorname{div}(J\boldsymbol{\sigma}F^{-T}) = J\mathbf{f},$$

which reads in weak formulation

$$(J\partial_t\mathbf{v}, \phi) + A(\kappa)(\mathbf{v}, \phi) - B(\kappa)(p, \phi) + B(\kappa)(\mathbf{v}, q) = (J\mathbf{f}, \phi) \quad \forall \phi, q \in \mathcal{V} \times \mathcal{Q} \quad (22)$$

where $B(\kappa)(\mathbf{v}, q) := (J\mathbf{F}^{-1}\mathbf{v}, \nabla q)$ and

$$A(\kappa)(\mathbf{v}, \phi) = (J(\kappa)\nabla\mathbf{v}\mathbf{F}(\kappa)^{-1}, \nabla\phi\mathbf{F}(\kappa)^{-1})_{\Omega}.$$

The second-order operator A is symmetric positive definite. Due to the regularity of the reference map (3), the map $\kappa \rightarrow A(\kappa)$ is differentiable. Its derivative (21) is bounded as long as contact of the boundary walls is prevented, i.e. as long as $\kappa \leq \kappa_{\max}$ is bound away from 1.5, compare (2) and (3). A diagonalisation of this system based on eigenvalues of the Stokes operator (in ALE mapping) allows for a reduction to a system of ordinary differential equations of type (20b), where $A(\kappa) \geq A_{\min} = \lambda_{\min} > 0$ can be estimated by the smallest eigenvalue. The results shown below could be obtained for the Stokes equations (22) as well. The restriction to the ODE (20a)-(20b) simplifies the presentation, however, significantly. The transfer of the arguments to a rigorous setting including nonlinear and non-symmetric equations is beyond the scope of this article.

Transferring the multiscale idea to this simplified system of ODEs gives the following system

$$\bar{\kappa}'(t) = \int_t^{t+1} R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) \, ds \quad \bar{\kappa}(0) = 0 \quad (23a)$$

$$\mathbf{v}^{\bar{\kappa}'}(t) + A(\bar{\kappa})\mathbf{v}^{\bar{\kappa}}(t) = f(t) \quad \mathbf{v}^{\bar{\kappa}}(t+1) = \mathbf{v}^{\bar{\kappa}}(t). \quad (23b)$$

We start with some preliminary results.

Lemma 5 (Periodic solutions). *Let $0 \leq \kappa \leq \kappa_{\max}$ be fixed. For the periodic solution*

$$\partial_t\mathbf{v}^{\kappa}(t) + A(\kappa)\mathbf{v}^{\kappa}(t) = f(t), \quad \mathbf{v}^{\kappa}(1) = \mathbf{v}^{\kappa}(0), \quad t \in [0, 1],$$

with $A(\kappa) \geq A = \min > 0$ it holds

$$|\mathbf{v}^{\kappa}(t)| \leq \frac{|f|_{L^{\infty}([0,1])}}{A(\kappa)}. \quad (24)$$

Further, for $0 \leq \kappa, \eta \leq \kappa_{max}$ let $\mathbf{v}^\kappa(t), \mathbf{v}^\eta(t)$ be two such periodic solutions. It holds

$$|\mathbf{v}^\kappa(t) - \mathbf{v}^\eta(t)| \leq \frac{|f|_{L^\infty[0,1]}}{A(\kappa)^2} |A(\kappa) - A(\eta)|. \quad (25)$$

Proof. (i) The solution is given by

$$\mathbf{v}^\kappa(t) = \exp(-A(\kappa)t) \left(\mathbf{v}^\kappa(0) + \int_0^t f(s) \exp(A(\kappa)s) ds \right). \quad (26)$$

For $t = 1$ we obtain using $\mathbf{v}^\kappa(1) = \mathbf{v}^\kappa(0)$

$$|\mathbf{v}^\kappa(0)| \left(1 - \exp(-A(\kappa)) \right) \leq \max_t |f(t)| \frac{1 - \exp(-A(\kappa))}{A(\kappa)} \Rightarrow |\mathbf{v}^\kappa(0)| = |\mathbf{v}^\kappa(1)| \leq \frac{|f|_\infty}{A(\kappa)}.$$

Inserting this estimate into (26), we get the estimate (24) for $t \in [0, 1]$.

(ii) Let $u(t) := \mathbf{v}^\kappa(t) - \mathbf{v}^\eta(t)$. It holds

$$u'(t) + A(\kappa)u(t) = (A(\eta) - A(\kappa))\mathbf{v}^\eta(t), \quad u(1) = u(0) = \mathbf{v}^\kappa(0) - \mathbf{v}^\eta(0).$$

Using the bound (24) we obtain

$$|u(t)| \leq \frac{|\mathbf{v}^\eta|_\infty}{A(\kappa)} |A(\eta) - A(\kappa)| \leq \frac{|f|_\infty}{A(\kappa)^2} |A(\eta) - A(\kappa)|.$$

□

Next, we show Lipschitz continuity of the reaction term $R(\cdot, \cdot)$.

Lemma 6 (Lipschitz continuity). *Let $\kappa, \eta, \mathbf{v}, \mathbf{u} \in C(I)$. It holds*

$$(i) \quad |R(\mathbf{v}, \kappa) - R(\mathbf{v}, \eta)| \leq \alpha |\kappa - \eta| \quad (27a)$$

$$(ii) \quad |R(\mathbf{v}, \kappa) - R(\mathbf{u}, \kappa)| \leq \alpha |\mathbf{u} - \mathbf{v}|. \quad (27b)$$

Proof. Given the definition of $R(\cdot, \cdot)$ in (20a) it holds

$$\begin{aligned} |R(\mathbf{v}, \kappa) - R(\mathbf{u}, \kappa)| &= \frac{|\mathbf{u} + \mathbf{v}|}{(1 + \kappa)(1 + \mathbf{u}^2)(1 + \mathbf{v}^2)} \alpha |\mathbf{v} - \mathbf{u}| \leq \alpha |\mathbf{v} - \mathbf{u}| \\ |R(\mathbf{v}, \kappa) - R(\mathbf{v}, \eta)| &= \frac{1}{(1 + \kappa)(1 + \eta)(1 + \mathbf{v}^2)} \alpha |\kappa - \eta| \leq \alpha |\kappa - \eta|. \end{aligned} \quad (28)$$

□

Now we are able to show an estimate between the dynamic solution $\mathbf{v}(t)$ defined by (20b) and the periodic solutions $\mathbf{v}^{\kappa(t)}(t)$ defined via (23b) for a given function $\kappa(t)$.

Lemma 7. *Let $\kappa \in C(I)$ with $\kappa(0) = 0$ and $0 \leq \kappa' \leq c_R \alpha$ be given. Further, let $\mathbf{v}(t)$ be the dynamic solution to (20b) for this given $\kappa(t)$ and the initial value $\mathbf{v}(0) = \mathbf{v}^0(0)$, i.e. the initial value of the periodic problem (23) to $\bar{\kappa} = 0$. Let $\mathbf{v}^{\kappa(t)}$ be the periodic solutions defined by (23b) for $\kappa = \kappa(t)$ at each time t . Further, let $A(\cdot)$ be differentiable and satisfy (21). It holds*

$$|\mathbf{v}(t) - \mathbf{v}^{\kappa(t)}(t)| \leq c(c_A, f, A_{\min}) \alpha.$$

Proof. Applying the chain rule, we see that the function $\mathbf{v}^{\kappa(t)}(t)$ fulfils the ODE

$$\partial_t \mathbf{v}^{\kappa(t)}(t) + \frac{d\mathbf{v}^{\kappa(t)}}{d\kappa(t)}(t) \kappa'(t) + A(\kappa(t)) \mathbf{v}^{\kappa(t)}(t) = 0, \quad \mathbf{v}^{\kappa(0)}(0) = \mathbf{v}(0).$$

Thus, it holds for the difference $w(t) := \mathbf{v}(t) - \mathbf{v}^{\kappa(t)}(t)$

$$\partial_t w(t) + A(\kappa(t)) w(t) = -\frac{d\mathbf{v}^{\kappa(t)}}{d\kappa(t)}(t) \kappa'(t), \quad w(0) = 0$$

with the solution

$$w(t) = -\int_0^t \frac{d\mathbf{v}^{\kappa(s)}}{d\kappa(s)}(s) \kappa'(s) \exp\left(-\int_s^t A(\kappa(r)) dr\right) ds. \quad (29)$$

To estimate the derivative $\frac{d\mathbf{v}^{\kappa(s)}}{d\kappa(s)}$ we consider the two time-periodic solution $\mathbf{v}^\kappa(t)$ and $\mathbf{v}^\eta(t)$ for fixed $0 \leq \kappa, \eta \leq \kappa_{max}$. We estimate their difference by (25) in Lemma 5

$$\frac{|\mathbf{v}^\kappa(t) - \mathbf{v}^\eta(t)|}{|\kappa - \eta|} \leq \frac{|f|_\infty}{A_{min}} \left| \frac{A(\kappa) - A(\eta)}{\kappa - \eta} \right|. \quad (30)$$

This bound is uniform in t and κ, η such that differentiability of $A(\kappa)$ (21) gives

$$\left| \frac{d\mathbf{v}^\kappa(t)}{d\kappa(t)}(t) \right| = \lim_{\eta \rightarrow \kappa} \frac{|\mathbf{v}^\kappa(t) - \mathbf{v}^\eta(t)|}{|\kappa - \eta|} \leq c_A \frac{|f|_\infty}{A_{min}} \leq c(c_A, f, A_{min}).$$

This allows to estimate (29) by

$$|w(t)| = |\mathbf{v}(t) - \mathbf{v}^{\kappa(t)}(t)| \leq c(c_A, f, A_{min}) \alpha.$$

□

In the next step, we study the different evolutions of $\kappa(t)$ governed by (20a) and $\bar{\kappa}(t)$ governed by (23a).

Lemma 8. *Let $(\kappa(t), \mathbf{v}(t))$ and $(\bar{\kappa}(t), \mathbf{v}^{\bar{\kappa}(t)}(t))$ be defined by (20a), (20b) and (23a), (23b), respectively with the initial values $\kappa(0) = \bar{\kappa}(0) = 0$ and $\mathbf{v}(0) = \mathbf{v}^0(0)$. For $0 \leq t \leq T = \mathcal{O}(\alpha^{-1})$ it holds*

$$|\bar{\kappa}(t) - \kappa(t)| \leq c(c_A, f, A_{min}) \alpha.$$

Proof. We introduce $w(t) := \bar{\kappa}(t) - \int_t^{t+1} \kappa(s) ds$. For $w_0 := w(0)$ it holds with (19) that $|w_0| = \mathcal{O}(\alpha)$. We insert $\pm R(\mathbf{v}(s), \bar{\kappa}(t))$

$$\begin{aligned} w'(t) &= \int_t^{t+1} \left(R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) - R(\mathbf{v}(s), \kappa(s)) \right) ds \\ &= \int_t^{t+1} \left(R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) - R(\mathbf{v}(s), \bar{\kappa}(t)) \right) + \left(R(\mathbf{v}(s), \bar{\kappa}(t)) - R(\mathbf{v}(s), \kappa(s)) \right) ds. \end{aligned} \quad (31)$$

Lipschitz continuity of $R(\cdot, \cdot)$ in Lemma 6 gives

$$|w'(t)| \leq \alpha \int_t^{t+1} |\mathbf{v}^{\bar{\kappa}(t)}(s) - \mathbf{v}(s)| ds + \alpha \int_t^{t+1} |\bar{\kappa}(t) - \kappa(s)| ds \quad (32)$$

The second term is estimated by inserting $\pm \int_t^{t+1} \kappa(r) dr$ and by using $\kappa' \leq \alpha$

$$\alpha \int_t^{t+1} |\bar{\kappa}(t) - \kappa(s)| ds \leq \alpha |w(t)| + \alpha \int_t^{t+1} \left| \int_t^{t+1} \kappa(r) - \kappa(s) dr \right| ds \leq \alpha |w(t)| + c\alpha^2. \quad (33)$$

To estimate the first term in (32) we introduce $\pm \mathbf{v}^{\kappa(s)}(s)$ and use Lemma 5 and Lemma 7

$$\begin{aligned} \alpha \int_t^{t+1} |\mathbf{v}^{\bar{\kappa}(t)}(s) - \mathbf{v}(s)| ds &\leq \alpha \int_t^{t+1} \left(|\mathbf{v}^{\bar{\kappa}(t)}(s) - \mathbf{v}^{\kappa(s)}(s)| + |\mathbf{v}^{\kappa(s)}(s) - \mathbf{v}(s)| \right) ds \\ &\leq c\alpha \int_t^{t+1} |\bar{\kappa}(t) - \kappa(s)| ds + c\alpha^2, \end{aligned} \quad (34)$$

where $c = c(c_A, f, A_{\min})$. An estimate for the remaining term is given in (33) and we combine (31)-(34) to find the relation

$$-(c\alpha^2 + c\alpha |w(t)|) \leq w'(t) \leq (c\alpha^2 + c\alpha |w(t)|).$$

An estimate for $|w(t)|$ follows by the solution of the corresponding ODE with initial value $w(0) = w_0$ and $|w_0| \leq c\alpha$

$$|w(t)| \leq c\alpha \exp(c\alpha t) \quad (35)$$

which satisfies $|w(t)| = \mathcal{O}(\alpha)$ for $t \leq T = \mathcal{O}(\alpha^{-1})$. Finally it holds with (35) and since $\kappa' \leq \alpha$

$$|\bar{\kappa}(t) - \kappa(t)| \leq \left| \bar{\kappa}(t) - \int_t^{t+1} \kappa(s) ds \right| + \left| \int_t^{t+1} \kappa(s) - \kappa(t) ds \right| \leq c\alpha.$$

□

4 Time discretisation

In this section we introduce second-order time-stepping schemes to approximate (23a)-(23b). The schemes are based on the second-order Adams-Bashforth scheme for the slow scale and a Crank-Nicolson scheme for the fast scale. Both choices are exemplarily and can in principle be substituted by any suitable time-stepping scheme. We choose an explicit scheme for the slow scale in order to avoid that several fast-scale problems have to be solved in each time step, see Remark 11 below. For better readability, we formulate the algorithm for the simplified system, although we will apply it to the full Navier-Stokes system in the numerical results section below.

4.1 Second-order multiscale schemes

First, we split the time interval $I = [0, T]$ into sub-intervals

$$0 = T_0 < T_1 < \dots < T_N, \quad K := T_n - T_{n-1},$$

that are for simplicity of uniform size. We define approximations $\bar{\kappa}_n := \bar{\kappa}(T_n)$ based on the second-order Adams-Bashforth multistep method

$$(AB) \quad \frac{\bar{\kappa}_{n+1} - \bar{\kappa}_n}{K} = \frac{3}{2} \int_0^1 R(\mathbf{v}_k^{\bar{\kappa}_n}, \bar{\kappa}_n) ds - \frac{1}{2} \int_0^1 R(\mathbf{v}_k^{\bar{\kappa}_{n-1}}, \bar{\kappa}_{n-1}) ds. \quad (36)$$

As the solutions $\mathbf{v}^{\bar{\kappa}_n}$ are periodic, we have shifted the interval $[t-1, t]$ to $I_P = [0, 1]$. In order to compute the required starting value $\bar{\kappa}_1$ for the Adams-Bashforth scheme, we start with one forward Euler step, which is sufficient to obtain second-order convergence

$$\frac{\bar{\kappa}_1 - \bar{\kappa}_0}{K} = \int_0^1 R(\mathbf{v}_k^{\bar{\kappa}_0}, \bar{\kappa}_0) ds. \quad (37)$$

These schemes are formally explicit, depend however on the averaged fast scale influences $R(\mathbf{v}_k^{\kappa_n}, \kappa_n)$. To compute these terms, we introduce a (for simplicity again uniform) temporal subdivision of the fast periodic interval $I_P = [0, 1]$ of step size k

$$0 = t_0 < t_1 < \dots < t_M = 1, \quad k := t_m - t_{m-1}.$$

For κ given, we introduce the notation $\mathbf{v}_m^\kappa := \mathbf{v}_k^\kappa(t_m)$ and approximate the periodic solution on the fast scale with the Crank-Nicolson time-stepping scheme

$$\begin{aligned} \frac{\mathbf{v}_m^\kappa - \mathbf{v}_{m-1}^\kappa}{k} + \frac{1}{2}A(\kappa)(\mathbf{v}_m^\kappa + \mathbf{v}_{m-1}^\kappa) &= \frac{1}{2}(f(t_m) + f(t_{m-1})), \quad m = 1, \dots, M, \\ |\mathbf{v}_M^\kappa - \mathbf{v}_0^\kappa| &\leq \epsilon_P. \end{aligned} \quad (38)$$

Based on the approximations made in the previous section, we introduce the following two-scale method:

Algorithm 9 (Explicit temporal two-scale method). *Given subdivisions $T_0 < T_1 < \dots < T_N$ of $I = [0, T]$ of step size K and $t_0 < t_1 < \dots < t_M$ of $I_P = [0, 1]$ of step size $k \ll K$. Let $\bar{\kappa}_0 = 0$. Iterate for $n = 0, 1, \dots, N$*

1. *For $\kappa = \bar{\kappa}_n$ solve the periodic problem (38) for $\mathbf{v}_m^{\bar{\kappa}_n}$, $m = 1, \dots, M$.*
2. *Compute the averaged feedback*

$$\bar{R}(\mathbf{v}_k^{\bar{\kappa}_n}, \bar{\kappa}_n) = \frac{\alpha k}{1 + \bar{\kappa}_n} \sum_{m=1}^M \frac{1}{1 + (\mathbf{v}_m^{\bar{\kappa}_n})^2}.$$

3. *Step forward $\bar{\kappa}_n \rightarrow \bar{\kappa}_{n+1}$ with (36).*

Remark 10. *In practice we ensure in Step 1 that the solution is periodic up to a certain threshold $\|\mathbf{v}_M^\kappa - \mathbf{v}_0^\kappa\| \leq \epsilon_P$. The box-rule used to compute the averaged wall shear stress in Step 2 of the algorithm is therefore equivalent to the second order trapezoidal rule (up to the small error $\mathcal{O}(k\epsilon_P)$).*

The main computational cost comes from the approximation of the periodic solutions $(\mathbf{v}_m^\kappa, p_m^\kappa)$ for a frozen value of κ . The efficient computation of these periodic problems is discussed below.

Remark 11 (Implicit multiscale schemes). *We are considering the rather simple interaction of the Navier-Stokes equations with a scalar ODE. For more detailed models, for example a boundary PDE to model the spatially diverse accumulation of $\kappa(x, t)$ along the boundary $\Gamma(\kappa)$ or even a full PDE-PDE model considering dynamical fluid-structure interactions and a detailed modelling of the bio-chemical processes causing plaque growth as introduced by Yang, Neuss-Radu et al. [39, 40], stiffness issues may call for implicit discretisations of the equation for κ . To realise an implicit multiscale method, e.g. based on the Crank-Nicolson scheme for both temporal scales an outer iteration must be introduced.*

4.2 Error analysis

We consider the system of equations (20a)-(20b), its multiscale approximation (23a)-(23b) and the discrete problem (36)-(38). Concerning the short-scale problem, we make the following assumption.

Assumption 12 (Approximation of the periodic problem). *Let $\epsilon_P > 0$ be the tolerance for reaching periodicity. We assume that there exists a constant $c > 0$ such that the Crank-Nicolson discretisation (38) to the flow problem satisfies the bound*

$$\|\mathbf{v}_M^\kappa - \mathbf{v}_0^\kappa\| + \|\mathbf{v}^\kappa(t_n) - \mathbf{v}_n^\kappa\| \leq c_{CN} k^2 + \epsilon_P,$$

for all $n = 1, \dots, M$ and where the constant c_{CN} does not depend on α and κ . It will however depend on the regularity of the fast variable $\mathbf{v}^\kappa(t)$.

Remark 13 (Approximation of the periodic problem). *Considering ODEs, the error estimate for the Crank-Nicolson scheme is standard and can be found in many textbooks. Applied to the Navier-Stokes equations optimal order estimates under realistic regularity assumptions are given in [19]. Similar estimates that also include second-order in time estimates for the pressure (which might be required for a stress-based feedback) are given in [35]. For algorithms to control the periodicity error, we refer to Section 4.3 below.*

Lemma 14 (Regularity of the solution). *Let $\bar{\kappa}$ be the solution to (23a) for $\mathbf{v}^{\bar{\kappa}} \in C(0, T)$ and $0 \leq \bar{\kappa} \leq \kappa_{\max}$. Moreover, let the map $\bar{\kappa} \rightarrow A(\bar{\kappa})$ be twice differentiable with bounded second derivatives. It holds*

$$\bar{\kappa} \in C^3(I), \quad \max_{[0, T]} |\bar{\kappa}''| = \mathcal{O}(\alpha^2), \quad \max_{[0, T]} |\bar{\kappa}'''| = \mathcal{O}(\alpha^3)$$

Proof. Let us first note, that $\bar{\kappa}'$ is bounded due to the continuity of the right-hand side $R(\mathbf{v}^{\bar{\kappa}(t)}, \bar{\kappa}(t))$ of (23a). Next, we consider the (total) temporal derivative of the right-hand side. The chain rule gives

$$d_t \int_0^1 R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) ds = R(\mathbf{v}^{\bar{\kappa}(t)}(1), \bar{\kappa}(t)) - R(\mathbf{v}^{\bar{\kappa}(t)}(0), \bar{\kappa}(t)) + \int_0^1 d_t R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) ds$$

The first part vanishes due to the periodicity of $\mathbf{v}^{\bar{\kappa}(t)}$.

For the second part we have

$$d_t R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) = -\frac{\alpha \kappa'(t)}{1 + \bar{\kappa}(t)^2} \int_0^1 \frac{1}{1 + (\mathbf{v}^{\bar{\kappa}(t)}(s))^2} ds - \frac{\alpha}{1 + \bar{\kappa}(t)} \int_0^1 \frac{2\mathbf{v}^{\bar{\kappa}(t)}(s) d_t \mathbf{v}^{\bar{\kappa}(t)}(s)}{(1 + (\mathbf{v}^{\bar{\kappa}(t)}(s))^2)^2} ds$$

As in the proof of Lemma 7 we show

$$d_t \mathbf{v}^{\bar{\kappa}(t)} = \frac{d\mathbf{v}^{\bar{\kappa}(t)}}{d\bar{\kappa}} \bar{\kappa}' = \mathcal{O}(\alpha).$$

In combination with the bound $\bar{\kappa}'(t) \leq c\alpha$, we obtain

$$|\bar{\kappa}''(t)| = \left| d_t \int_0^1 R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) ds \right| \leq c\alpha^2.$$

A similar argumentation yields for the third derivative

$$|\bar{\kappa}'''(t)| = \left| d_t^2 \int_0^1 R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) ds \right| = \left| \int_0^1 d_t^2 R(\mathbf{v}^{\bar{\kappa}(t)}(s), \bar{\kappa}(t)) ds \right| \leq c\alpha^3,$$

where we have used that

$$d_t^2 \mathbf{v}^{\bar{\kappa}(t)} = \frac{d\mathbf{v}^{\bar{\kappa}(t)}}{d\bar{\kappa}} \bar{\kappa}'' + \frac{d^2 \mathbf{v}^{\bar{\kappa}(t)}}{d\bar{\kappa}^2} \bar{\kappa}'^2 = \mathcal{O}(\alpha^2),$$

given that $A(\kappa)$ is twice differentiable in κ . $\square$

Lemma 15 (Local approximation error of the effective equation). *Let $\bar{\kappa} \in C^3(I)$ be the solution to (23), $\bar{\kappa}_K \in \mathbb{R}^{N+1}$ the approximation by (36), where the periodic velocity problems are approximated by (38).*

For $T_n \leq c\alpha^{-1}$, the error $e_n := \bar{\kappa}(T_n) - \bar{\kappa}_n$ between the solution $\bar{\kappa}$ of (23a) and the iterates $\bar{\kappa}_n$ of the Adams-Bashforth discretisation (36) is bounded by

$$|e_n| = C(\alpha^2 K^2 + k^2 + \epsilon_P)$$

with a generic constant C .

Proof. Combination of Taylor expansions around T_n and T_{n-1} gives for the continuous solution $\bar{\kappa}$

$$\begin{aligned} \bar{\kappa}(T_{n+1}) &= \bar{\kappa}(T_n) + \frac{3K}{2} \int_0^1 R(\mathbf{v}^{\bar{\kappa}(T_n)}(s), \bar{\kappa}(T_n)) ds \\ &\quad - \frac{K}{2} \int_0^1 R(\mathbf{v}^{\bar{\kappa}(T_{n-1})}(s), \bar{\kappa}(T_{n-1})) ds + \mathcal{O}(K^3) \max_{\xi \in (T_{n-1}, T_{n+1})} |\kappa'''(\xi)|. \end{aligned}$$

In combination with (36), we obtain the error representation

$$\begin{aligned} e_{n+1} &= e_n + \frac{3K}{2} \int_0^1 \left(R(\mathbf{v}^{\bar{\kappa}(T_n)}(s), \bar{\kappa}(T_n)) - R(\mathbf{v}_{k,n}^{\bar{\kappa}_n}(s), \bar{\kappa}_n) \right) ds \\ &\quad - \frac{K}{2} \int_0^1 \left(R(\mathbf{v}^{\bar{\kappa}(T_{n-1})}(s), \bar{\kappa}(T_{n-1})) - R(\mathbf{v}_k^{\bar{\kappa}_{n-1}}(s), \bar{\kappa}_{n-1}) \right) ds + \mathcal{O}(K^3) \max_{\xi \in (T_{n-1}, T_{n+1})} |\kappa'''(\xi)| \end{aligned}$$

By means of Lemma 6, we can estimate

$$\int_0^1 |R(\mathbf{v}^{\bar{\kappa}(T_n)}(s), \bar{\kappa}(T_n)) - R(\mathbf{v}_k^{\bar{\kappa}_n}(s), \bar{\kappa}_n)| ds \leq \alpha \left(\int_0^1 |\mathbf{v}^{\bar{\kappa}(T_n)}(s) - \mathbf{v}_k^{\bar{\kappa}_n}(s)| ds + |\bar{\kappa}(T_n) - \bar{\kappa}_n| \right).$$

For the first part, we use the estimate (25) of Lemma 5 and Assumption 12

$$\begin{aligned} \int_0^1 |\mathbf{v}^{\bar{\kappa}(T_n)}(s) - \mathbf{v}_k^{\bar{\kappa}_n}(s)| ds &\leq \int_0^1 |\mathbf{v}^{\bar{\kappa}(T_n)} - \mathbf{v}^{\bar{\kappa}_n}| ds + \int_0^1 |\mathbf{v}^{\bar{\kappa}_n} - \mathbf{v}_k^{\bar{\kappa}_n}| ds \\ &\leq c_A \frac{|f|_{L^\infty[0,1]}}{A_{\min}^2} |\bar{\kappa}(T_n) - \bar{\kappa}_n| + c_{CN} k^2 \|\mathbf{v}^{\bar{\kappa}_n}\|_{C^3([0,1])} + \epsilon_P. \end{aligned}$$

In combination with Lemma 14 this yields

$$|e_{n+1}| \leq |e_n| + c\alpha K (|e_n| + |e_{n-1}| + k^2 + \epsilon_P) + c\alpha^3 K^3.$$

Summing over $n = 1, \dots, N-1$ and using $e_0 = 0$, we obtain

$$|e_N| \leq |e_1| + c\alpha T_N (k^2 + \epsilon_P + \alpha^2 K^2) + \sum_{n=1}^{N-1} c\alpha K |e_n|.$$

The error e_1 depends on the forward Euler method, that is used to compute $\bar{\kappa}_{K,1}$

$$|e_1| \leq cK^2|\kappa''|_\infty \leq c\alpha^2K^2,$$

where we have used Lemma 14 and $|e_0| = 0$. Finally, a discrete Gronwall inequality yields

$$|e_N| \leq c\alpha T_N \left(k^2 + \alpha^2 K^2 + \epsilon_P \right) \exp \left(\alpha T_N \right).$$

The postulated result follows for $T_N = \mathcal{O}(\alpha^{-1})$. □

Finally, we can estimate the error between the multiscale algorithm and the solution $\kappa(t)$ to the original coupled problem.

Theorem 16 (A priori estimate for the multiscale algorithm). *Let $I = [0, T]$ with $T = \mathcal{O}(\alpha^{-1})$ and let $\kappa \in C(I)$ and $\bar{\kappa}_K$ be the solutions to the original problem (1) and the discrete effective equations (36), respectively. It holds*

$$|\kappa(T_n) - \bar{\kappa}_n| = c \left(k^2 + \alpha^2 K^2 + \epsilon_P + \alpha \right),$$

where $c > 0$ does not depend on α, K, k and ϵ_P .

Proof. We introduce $\pm \bar{\kappa}(T_n)$ and estimate

$$|\kappa(T_n) - \bar{\kappa}_n| \leq |\kappa(T_n) - \bar{\kappa}(T_n)| + |\bar{\kappa}(T_n) - \bar{\kappa}_n|.$$

The two terms on the right hand side are estimated with Lemma 8 and Lemma 15. □

4.3 Approximation of the periodic flow problem

The temporal multiscale schemes are based on periodic solutions $\mathbf{v}^{\bar{\kappa}_n}(s)$ for $s \in [0, 1]$, where the variable $\bar{\kappa}_n = \bar{\kappa}(t_n)$ is frozen such that no feedback between fluid problem and reaction equation takes place within this short interval. A numerical difficulty lies in the determination of the correct initial value $\mathbf{v}^{\bar{\kappa}_n}(0) = \mathbf{v}_0^{\bar{\kappa}_n}$. Let us consider again the full Navier-Stokes problem

$$\nabla \cdot \mathbf{v}^{\bar{\kappa}_n} = 0, \quad \rho(\partial_t \mathbf{v}^{\bar{\kappa}_n} + (\mathbf{v}^{\bar{\kappa}_n} \cdot \nabla) \mathbf{v}^{\bar{\kappa}_n}) - \operatorname{div} \boldsymbol{\sigma}(\mathbf{v}^{\bar{\kappa}_n}, p^{\bar{\kappa}_n}) = \mathbf{f}, \quad \mathbf{v}^{\bar{\kappa}_n}(1) = \mathbf{v}^{\bar{\kappa}_n}(0) \text{ in } \Omega(\kappa_n).$$

We assume that such a periodic solution of the Navier-Stokes equations exists. Some results are given by Kyed and Galdi [13, 14, 25] that require, however, severe restrictions on the problem data. Depending on the transient dynamics, the decay of the nonstationary solution to this periodic solution can be very slow. It depends basically on $\exp(-\nu\lambda T)$, where ν is the viscosity and λ the smallest eigenvalue of the Stokes operator, which depends on the inverse of the domain diameter. Several acceleration techniques that are based on shooting methods [15, 43], Newton schemes [36, 23] or gradient-based optimisation techniques [17, 33] have been proposed to quickly identify the initial value $\mathbf{v}_0^{\bar{\kappa}_n}$. For the specific problem setup used in this work a few forward iterations were sufficient to reach a near-periodic state.

5 Numerical examples

We consider the full problem described in the introduction, namely the incompressible Navier-Stokes equations coupled to a scalar ODE model. In order to transfer the proofs from the simplified setting to this more relevant case, several open questions regarding the existence and regularity theory of the Navier-Stokes equations in the periodic setting would have to be addressed. Even if these large open questions are avoided by suitable assumptions, the multiscale analysis is significantly complicated by the problem of non-uniqueness of periodic solutions. The numerical results presented in the section will, however, reveal convergence rates and also constants that are in full agreement with the simplified theory.

5.1 Configuration of the model problem

We start by specifying all parameters of a typical test-case. The test case is inspired by the long-term formation of atherosclerotic plaques in blood vessels [40, 6] and although we strongly simplify the underlying models we aim at matching the typical spatial and temporal scales of this problem. The parameters are tuned in such a way that $T_{long} = \mathcal{O}(1 \text{ month})$ and $t_{fast} = 1 \text{ s}$ is the period of the blood flow. The flow domain is given in (2), see also Figure 1.

We estimate an average inflow of approximately 45 cm^3 in every heart beat with a period of $P_{fast} = 1 \text{ s}$. We use a simple periodic Dirichlet inflow profile

$$\mathbf{v}_{in}(y, t) = 25 \sin(\pi t)^2 \left(1 - \frac{y^2}{1.5^2}\right) \text{ cm/s} \quad \text{on } \Gamma_{in} \times [0, T]. \quad (39)$$

A maximum velocity of 25 cm/s is reached. On the outflow boundary Γ_{out} we specify the *do-nothing* outflow condition

$$\rho \nu \partial_{\vec{n}} \mathbf{v} - p \vec{n} = 0 \quad (40)$$

that includes a pressure normalising condition $\int_{\Gamma_{out}} p \, d\mathbf{o} = 0$, see [20]. Kinematic viscosity and density resemble blood and the parameters in the reaction term (5) are tuned to obtain a realistic behaviour concerning the different temporal scales of atherosclerotic plaque growth

$$\rho = 1 \text{ g/cm}^3, \quad \nu = 0.04 \text{ cm}^2 \cdot \text{s}^{-1}, \quad \alpha = 5 \cdot 10^{-5}, \quad \sigma_0 = 30. \quad (41)$$

The constant σ_0 is such that the concentration κ reaches the value 1 at approximately $T = \mathcal{O}(\alpha^{-1})$, i.e.

$$0 \leq R(\mathbf{v}, \kappa) = \frac{\alpha}{(1 + \kappa)(1 + \sigma_{WSS}(\mathbf{v})^2)} \leq c_R \alpha$$

with a constant $c_R > 0$ such that

$$\forall T \in [0, c\alpha^{-1}] : \quad 0 \leq \kappa(t) \leq c_R \text{ and } 0 \leq \kappa'(t) \leq c_R \alpha. \quad (42)$$

We exploit the symmetry of the problem and compute on the upper half of the domain only. On the symmetry boundary at $y = 0$ we prescribe the condition

$$\mathbf{v} \cdot \vec{n} = 0, \quad \boldsymbol{\sigma}(\mathbf{v}, p) \vec{n} \cdot \vec{\tau} = 0.$$

To cope with the moving domain character of this coupled problem we formulate the Navier Stokes equations in ALE coordinates [11, 32] using the map (3). The Navier-Stokes equations, mapped to the reference framework $\hat{\Omega} := \Omega(0)$ read

$$\begin{aligned} \mathbf{v}(0) = \mathbf{v}_0, \quad \text{div}(J\mathbf{F}^{-1}\mathbf{v}) = 0, \quad \rho J(\partial_t \mathbf{v} + (\mathbf{F}^{-1}(\mathbf{v} - \partial_t T) \cdot \nabla)\mathbf{v}) - \text{div}(J\mathbf{F}^{-T}\hat{\boldsymbol{\sigma}}(\mathbf{v}, p)\mathbf{F}^{-1}) = 0 \quad \text{in } \Omega \\ \hat{\boldsymbol{\sigma}}(\mathbf{v}, p) = -pI + \rho_f \nu_f \nabla \mathbf{v} \mathbf{F}^{-1}. \end{aligned} \quad (43)$$

In order to abbreviate the notation we introduce the differential operators

$$A(\kappa; \mathbf{v}, p) := \rho(\mathbf{F}^{-1}(\mathbf{v} - \partial_t T) \cdot \nabla) \mathbf{v} - \operatorname{div} \left(J \mathbf{F}^{-T} \hat{\boldsymbol{\sigma}}(\mathbf{v}, p) \mathbf{F}^{-1} \right), \quad B(\kappa; \mathbf{v}) := \operatorname{div} (J \mathbf{F}^{-1} \mathbf{v}) = 0,$$

where $\mathbf{F} = \mathbf{F}(\kappa)$, $J = J(\kappa)$ and $T = T(\kappa)$. The right-hand side $\mathbf{f}$ is zero in all examples.

5.2 Discretisation

We briefly sketch the discretisation in space and time. All numerical experiments have been realised in the software library Gascoigne 3D [4]. We use uniform time-steps k and K on both scales and the time-stepping schemes presented in (36)-(38).

For spatial discretisation we triangulate the reference domain $\hat{\Omega}$ into open quadrilaterals, allowing for local refinement based on hanging nodes, see [31] for details on the realisation in the software Gascoigne 3d. Equal-order biquadratic finite elements are used for velocity and pressure degrees of freedom. Pressure stabilisation is accomplished with the local projection stabilisation scheme [3]. Stabilisation of the convective terms is not required in the settings considered here.

5.2.1 Direct simulation

The PDE-ODE system is a two-way coupled problem. We will compare the presented two-scale scheme with a direct forward simulation. As we do not expect any stiffness-related problems in the ODE we decouple the PDE-ODE system by an implicit/explicit approach where, as in the two-scale approach, the discretisation of the Navier-Stokes equation is based on the second-order Crank-Nicolson scheme and the discretisation of the ODE on the second-order explicit Adams-Bashforth formula

$$\begin{aligned} \bar{\kappa}_{n+1} - \bar{\kappa}_n &= \frac{3k}{2} R(\mathbf{v}_n, \bar{\kappa}_n) - \frac{k}{2} R(\mathbf{v}_{n-1}, \bar{\kappa}_{n-1}), \\ \rho \bar{J}_{n+1}(\mathbf{v}_{n+1} - \mathbf{v}_n) &= \frac{k}{2} A(\bar{\kappa}_{n+1}; \mathbf{v}_{n+1}, p_{n+1}) + \frac{k}{2} A(\bar{\kappa}_n; \mathbf{v}_n, p_{n+1}), \quad B(\bar{\kappa}_{n+1}, \mathbf{v}_{n+1}) = 0, \end{aligned}$$

resulting in a two-step method that splits naturally into one explicit ODE-step and an implicit Navier-Stokes step.

5.3 Numerical results

5.3.1 Configuration with resolvable time scales

In a first test we take the value $\alpha = 5 \cdot 10^{-5}$ in (41). By this choice, the concentration κ reaches approximately 1 after about 50 000s such that we can still resolve the coupled problem in all temporal scales (although the direct simulation still takes a substantial effort). To keep the computational effort within bounds we use a rather coarse spatial discretisation with 320 elements, resulting in 4 131 unknowns of a biquadratic equal-order discretisation for velocities and pressure.

In Figure 3 (left) we give an overview of the temporal evolution of the concentration variable $\kappa(t)$ using a full resolution of the fast scale. The simulations break down at $t \approx 55\,000$ s due to the deterioration of the ALE map. For the small interval [10 000s, 10 002s] we show a close-up view of the resolved solution $\kappa_k(t)$ and the averaged value $\int_t^{t+1} \kappa_k(s) ds$. The deviation is bound by $3 \cdot 10^{-6}$, which is $\mathcal{O}(\alpha)$, in agreement with Lemma 8. We determine reference values $\kappa_{ref}(t_n)$ using the three different time-steps $k = 0.05$ s, $k = 0.025$ s and $k = 0.0125$ s and by extrapolation of the numerical results in all instants $t_n = n \cdot 0.05$ s that are resolved on all three time meshes. The relative errors $|\kappa_k(t_n) - \kappa_{ref}(t_n)|/|\kappa_{ref}(t_n)|$ (based on these extrapolated errors) are given in Figure 3 (right). Convergence rate

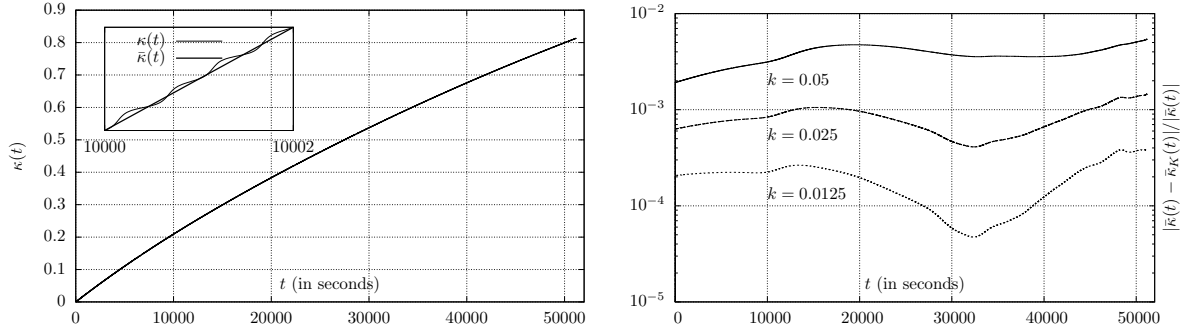


Figure 3: *Left*: Evolution of the concentration variable $\bar{\kappa}(t)$ as function over time (forward simulation with $k = 0.05s$). In the small subplot we show both $\bar{\kappa}(t)$ and the resolved variable $\kappa(t)$. The maximum deviation is bound by $3 \cdot 10^6$. *Right*: Relative error in $\bar{\kappa}(t)$ under refinement of the time step k (compared to extrapolated values).

in terms of k is approximately quadratic. Further, there is no significant accumulation of simulation errors over time.

By the evolution of the concentration κ over time, the computational domain undergoes substantial deformations with a strong narrowing of the flow domain. In Figure 4 we show snapshots of the solution at different time steps, $t \approx 6250s, 13500s, 18750s, \dots, 50000s$. The narrowing of the gap causes an acceleration of the fluid resulting in a higher Reynolds number flow with a substantial variation in the feedback functional $R(\mathbf{v}, \kappa)$ which depends on the wall shear stress.

5.3.2 Two-scale approach

Next, in Figure 5 we show the results obtained with the multiscale method. The tolerance for approximating the periodic flow problems is set to

$$\|\mathbf{v}^\kappa(1) - \mathbf{v}^\kappa(0)\|_{L^2(\Omega)}^2 + \|p^\kappa(1) - p^\kappa(0)\|_{L^2(\Omega)}^2 < \epsilon_P^2 := 10^{-8}.$$

For each of the three short time step sizes $k = 0.05s$, $k = 0.025s$ and $k = 0.0125s$ we use long time step sizes ranging from $K = 6400s$ to $K = 400s$. In the left plot we compare the results for different values of the long time step size K . In this (non-logarithmic) plot we see convergence of the results to the corresponding resolved simulation with the same short step size $k = 0.025s$. The lower plot shows the corresponding results for a variation of the small step size k , while the long scale step size is fixed to $K = 400s$. For comparison we show the results obtained with the resolved simulation for these small-step sizes. Again we see convergence of the two-scale scheme towards the resolved scheme.

The effect of the small step size k is dominant. This is highlighted by a closer analysis of the convergence at time $t = 51200s$, the results being shown in Table 1. We show the concentration and the errors for the different two-scale approaches as well as for the resolved forward simulation. We fit all these values to the postulated relation

$$\bar{\kappa}(k, K) = \bar{\kappa} + C_k k^{q_k} + C_K K^{q_K} \quad (44)$$

to get a better understanding of the convergence rates. We estimate all parameters $\kappa, C_k, C_K, q_k, q_K$ (obtained with gnuplot fit [38]) and find

$$\bar{\kappa}(k, K) = \bar{\kappa} - 1.12 \cdot k^{1.85} - 6.61 \cdot 10^{-10} \cdot K^{1.80},$$

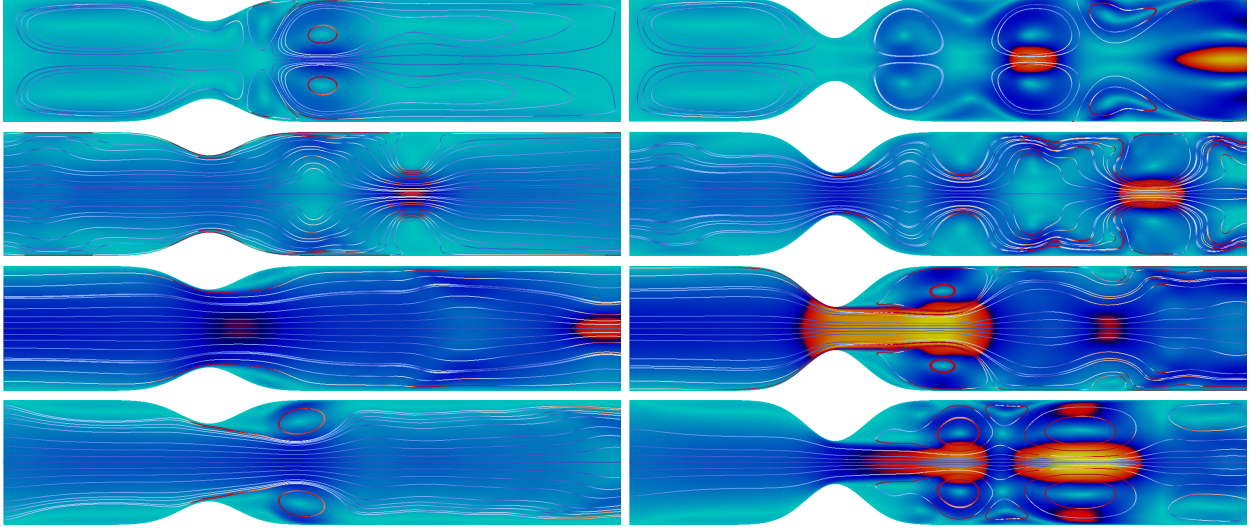


Figure 4: Velocity magnitude at times $t = n \cdot 6250\text{s}$ for $n = 1, 2, \dots, 8$ on domains with different growth. As the inflow profile is periodic, the narrowing of the domain causes a significant change of the flow pattern.

see also Table 1. Convergence is close to the expected second order. The most striking result is the variance in the constants that differ about α^2 . This result is in good correspondence to the error estimate derived in Theorem 16 where the constant in front of the K^2 -term depends on α^2 . Balanced discretisation errors are given for $\alpha K \approx k$, i.e. for $K \approx \alpha^{-1}k$.

Based on the time step relation we can compute the possible speedup of the two-scale approach which we measure in the overall number of Navier-Stokes time steps to be performed. The forward algorithm requires $E_{fwd} = \frac{T}{k}$ solution steps, while the two-scale approach has an effort of $E_{ms} = T/K \cdot n_{period}^{1/k} = T n_{period}/kK$ steps, where n_{period} is the number of cycles that are necessary to compute a periodic solution. Given $K \approx \alpha^{-1}k$ we approximate $E_{ms} \approx \alpha T n_{period}/k^2$, and the speedup is estimated by

$$\frac{E_{fwd}}{E_{ms}} = \frac{k}{\alpha n_{period}}.$$

In our example we have $n_{period} \approx 5$ and $\alpha = 5 \cdot 10^{-5}$ expecting a speedup of $4000k$. In Figure 6 we plot the error over the required number of Navier-Stokes time steps. By circles we indicate the two-scale results with a balanced error contribution, which we define as the state, where the error of the two-scale approach is within 10% of the error of a fully resolved simulation for the same k . We observe speedups of 250 for $k = 0.05\text{s}$, 180 for $k = 0.025\text{s}$ and 90 for $k = 0.0125\text{s}$, slightly better values than the predicted ones based on $4000k$. The overall computational time for the forward simulation with $k = 0.0125\text{s}$ was about 13 days, while the two-scale simulation with $k = 0.0125\text{s}$ and $K = 800\text{s}$, giving a comparable accuracy, was about 45 min.

5.3.3 Configuration with realistic time scales

Finally, we consider the coupled problem with the time scale parameter $\alpha = 10^{-6}$, which is close to the temporal dynamics of atherosclerotic plaque growth and 50 times smaller than in the first example. Here, a resolved forward simulation is not feasible. The concentration $\kappa(t)$ will reach a value of approximately 0.8 at $T \approx 2.5 \cdot 10^6\text{s} \approx 30$ days.

k	$\bar{\kappa}_K$			error (w.r.t. extrapolation)		
	0.05s	0.025s	0.0125s	0.05s	0.025s	0.0125s
$K = 6400\text{s}$	0.8089225	0.8118418	0.8126760	$1.11 \cdot 10^{-2}$	$7.55 \cdot 10^{-3}$	$6.53 \cdot 10^{-3}$
$K = 3200\text{s}$	0.8122441	0.8153539	0.8162696	$7.05 \cdot 10^{-3}$	$3.25 \cdot 10^{-3}$	$2.13 \cdot 10^{-3}$
$K = 1600\text{s}$	0.8132730	0.8164325	0.8173319	$5.80 \cdot 10^{-3}$	$1.93 \cdot 10^{-3}$	$8.35 \cdot 10^{-4}$
$K = 800\text{s}$	0.8135139	0.8166886	0.8175426	$5.50 \cdot 10^{-3}$	$1.62 \cdot 10^{-3}$	$5.77 \cdot 10^{-4}$
$K = 400\text{s}$	0.8135782	0.8167926	0.8176490	$5.42 \cdot 10^{-3}$	$1.49 \cdot 10^{-3}$	$4.47 \cdot 10^{-4}$
resolved	0.8135999	0.8168226	0.8176928	$5.42 \cdot 10^{-3}$	$1.46 \cdot 10^{-3}$	$3.93 \cdot 10^{-4}$

$$\begin{aligned} \text{Fit to } \bar{\kappa}_K(k, K) &= \bar{\kappa} + C_k k^{q_k} + C_K K^{q_K} & \bar{\kappa} &= 0.818006 \pm 10^{-3}\% \\ C_k &= -1.12 \pm 17\%, & C_k &= -6.61 \cdot 10^{-10} \pm 34\%, & q_k &= 1.85 \pm 3.39\%, & q_K &= 1.80 \pm 2.14\% \end{aligned}$$

Table 1: Convergence of the two-scale method at time $t = 51\,200\text{s}$. We show the values of $\bar{\kappa}_K$ and the error (w.r.t. the extrapolation in $k \rightarrow 0$ and $K \rightarrow 0$). We compare the results of the two-scale method with the fully resolved forward computation. Finally, we fit the numerical results to the expected convergence behaviour.

	$h = 0.16\text{cm}$			$h = 0.08\text{cm}$			$h = 0.04\text{cm}$		
	$k = 0.05\text{s}$	$k = 0.025\text{s}$	$k = 0.0125\text{s}$	$k = 0.05\text{s}$	$k = 0.025\text{s}$	$k = 0.0125\text{s}$	$k = 0.05\text{s}$	$k = 0.025\text{s}$	$k = 0.0125\text{s}$
$K = 204800\text{s}$	$4.45 \cdot 10^{-3}$	$2.46 \cdot 10^{-3}$	$2.20 \cdot 10^{-3}$	$4.06 \cdot 10^{-3}$	$2.42 \cdot 10^{-3}$	$2.04 \cdot 10^{-3}$	$4.06 \cdot 10^{-3}$	$2.42 \cdot 10^{-3}$	$2.04 \cdot 10^{-3}$
$K = 102400\text{s}$	$2.86 \cdot 10^{-3}$	$8.94 \cdot 10^{-4}$	$6.17 \cdot 10^{-4}$	$2.75 \cdot 10^{-3}$	$1.06 \cdot 10^{-3}$	$6.28 \cdot 10^{-4}$	$2.75 \cdot 10^{-3}$	$1.06 \cdot 10^{-3}$	$6.28 \cdot 10^{-4}$
$K = 51200\text{s}$	$2.43 \cdot 10^{-3}$	$4.76 \cdot 10^{-4}$	$1.96 \cdot 10^{-4}$	$2.39 \cdot 10^{-3}$	$7.04 \cdot 10^{-4}$	$2.58 \cdot 10^{-4}$	$2.39 \cdot 10^{-3}$	$7.04 \cdot 10^{-4}$	$2.58 \cdot 10^{-4}$
extrapolated ($k, K \rightarrow 0$)	$ \bar{\kappa}(\tilde{T}) - \bar{\kappa}_h(\tilde{T}) \approx 4.55 \cdot 10^{-2}$			$ \bar{\kappa}(\tilde{T}) - \bar{\kappa}_h(\tilde{T}) \approx 8.61 \cdot 10^{-3}$			$ \bar{\kappa}(\tilde{T}) - \bar{\kappa}_h(\tilde{T}) \approx 1.63 \cdot 10^{-3}$		

Table 2: Convergence of the two-scale approach for $\alpha = 10^{-6}$. On three mesh levels we indicate the errors in the concentration $\bar{\kappa}(\tilde{T})$ at $\tilde{T} = 1\,843\,200\text{s} \approx 21\text{ days}$. In each block, the error are given w.r.t. the extrapolation $k, K \rightarrow 0$. In the last line we indicate the (dominating) spatial error for each block.

Assuming the validity of the estimate (44) and in addition that $C_K \approx \alpha^2$ we expect balanced error contributions for $K \approx \alpha^{-1}k = 10^6k$. The character of the short scale problem does not depend on α . Hence we consider again the step sizes $k = 0.05\text{s}$, $k = 0.025\text{s}$ and $k = 0.0125\text{s}$. The large time step, however, can be significantly increased. We present the results for $\tilde{T} \approx 21\text{ days}$ in Table 2. For this second example, we vary also the mesh size h to discuss the impact of all relevant discretisation parameters. While a smaller value of α makes the time scale challenge more severe, the two-scale approach will profit, as the potential speedup will benefit from the relation $K \approx \alpha^{-1}k$.

Combining all 27 computations based on three values for h, k and K we find the relation

$$\bar{\kappa}_{h,k,K}(\tilde{T}) \approx 0.59076 + 7.6h^{2.4} - 1.7k^{2.2} - 0.04\alpha^2K^{1.9},$$

which shows approximately second order convergence in both time step sizes and the mesh size and also the proper scaling of the constants in the $\mathcal{O}(k^2)$ and $\mathcal{O}(K^2)$ terms. Spatial and temporal errors show a different sign which is also seen in Figure 7, where we show the errors for all computations. In the right sketch of this figure we compare the computational times of the two-scale approach with a hypothetical resolved simulation. Here, the errors are predicted by extrapolation. The computational times are based on the number of Navier Stokes steps, namely $k^{-1}\tilde{T}$ and the average computational time for each Navier Stokes step, which is 0.135s on the $h = 0.16\text{cm}$ mesh, 0.62s for $h = 0.08\text{cm}$ and

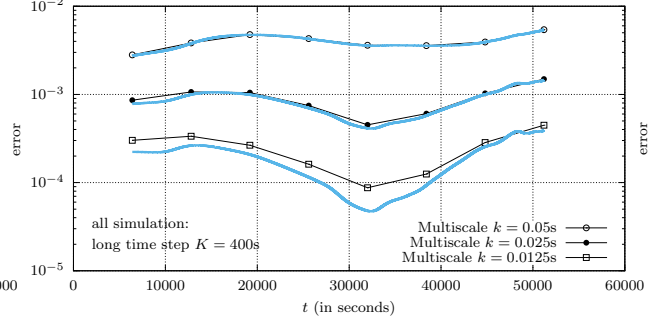
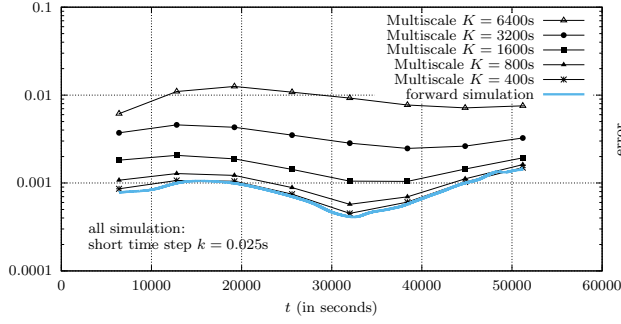


Figure 5: Convergence of the temporal two-scale method. *Left*: Effect of the long time step K using the small time step $k = 0.025s$. *Right*: Effect of the short time step k using the long time step size $K = 400s$. For comparison we plot the error of the fully resolved simulation using these time-step sizes.

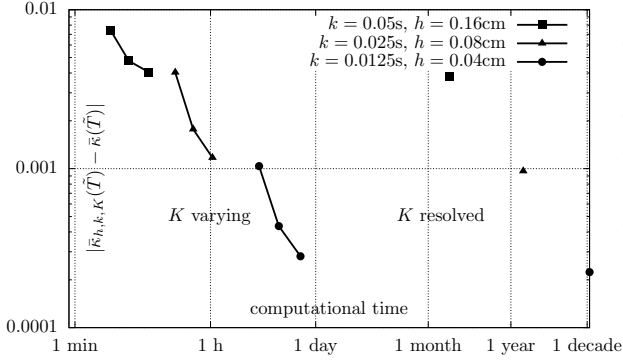


Figure 6: Computational effort (measured in Navier-Stokes time steps) for the two-scale approach (lines) and the resolved forward simulation (points). We use three small time steps k from 0.05s down to 0.0125s and vary the long time step K from 6400s to 400s. Circles indicate two-scale solutions of a quality comparable to the resolved forward simulation (at most 10% additional error). The computational time for one Navier Stokes step is about 0.2s (Core i7-7700, 3.60GHz, 1370 spatial unknowns, biquadratic finite elements).

2.3s for $h = 0.04cm$. The results are very similar to those shown in Figure 6 for the first example. The best two-scale results are close to the hypothetical resolved results. Here however, the savings are substantially larger, with 10 minutes vs. 2 months (factor 8000) for $h = 0.16cm$, 1 hour vs. nearly 2 years for $h = 0.08cm$ (factor 12000) and 15 hours vs. more than 10 years for $h = 0.04cm$ (factor 6000).

Finally, we also evaluate the effect of the parameter ϵ_P used to control the periodicity of the Navier-Stokes solution, compare Theorem 16. In Table 3 we show the errors at $\bar{T} = 1843200s$ for computations based on $K = 25600s$, $k = 0.0125s$ and $h = 0.08cm$. The effect of ϵ_P is very small.

6 Conclusion

We have presented a framework for the simulation of temporal multiscale problems, where we are interested in the evolution of a slow variable which depends on an oscillating fast variable. The numerical schemes are designed for underlying models that are given by partial differential equations. The most important assumption is a local (in time) proximity of the fast scale variable to the solution of a periodic problem. An effective scheme for the slow variable is derived by replacing the fast variable with the periodic solution which can be computed locally, as no initial values must be transferred. The only overhead of the multiscale scheme comes from the identification of initial values required for

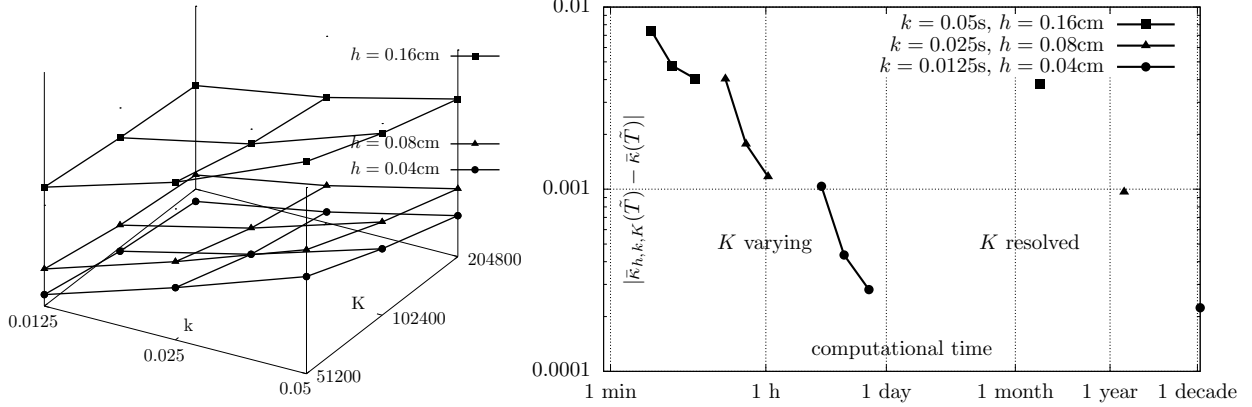


Figure 7: *Left*: Error of the multiscale method under refinement in h, k and K versus the extrapolated reference value. *Right*: Comparison of the computational times of the multiscale method with the corresponding results for a resolved forward simulation. These results are based on an extrapolation of the error and a prediction of the computational times by multiplying the number of required time steps with the average computing times for each step.

ϵ_P	10^{-1}	10^{-2}	10^{-3}	10^{-4}
$ \bar{\kappa}_{h,k,K}(T) _{\epsilon_P} - \bar{\kappa}_{h,k,K}(\tilde{T}) _{\epsilon_P=10^{-8}}$	$1.99 \cdot 10^{-5}$	$4.88 \cdot 10^{-7}$	$3.19 \cdot 10^{-7}$	$1.22 \cdot 10^{-7}$

Table 3: Impact of the periodicity parameter ϵ_P on the error in concentration $\bar{\kappa}$ in $\tilde{T} = 1843200$ s. Computed with respect to $\epsilon_P = 10^{-8}$. The discretisation is chosen as $h = 0.08$ cm, $k = 0.0125$ s and $K = 25600$ s.

approximating the periodic problems. Nevertheless, we gain huge speedups compared to a simulation with resolved time scales. The efficiency of the multiscale approach increases when the time scale separation gets larger ($\alpha \rightarrow 0$ in our context).

The resulting scheme depends on several numerical parameters, small and large time steps k and K , and the spatial mesh size h . It remains a topic for a future work to design an automatic and adaptive algorithm to control all these parameters in order to balance all contributing error terms.

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